Assignment No. 10: Curve Fitting by the Method of Least Squares

Blue = data from question; black = formulas. Σ in column K. Normal equations solved directly (no regression shortcuts).

Q1. Fit y = a + bx										Σ
x	0.	1.	2.	3.	4.					10.
y	1.	1.8	3.3	4.5	6.3					16.9
x·y	0.00	1.80	6.60	13.50	25.20					47.10
x²	0.00	1.00	4.00	9.00	16.00					30.00
Results	n	5	b	1.3300	a	0.7200	y = 0.7200 + 1.3300x			

Q2. Fit y = a + bx; estimate y at x = 72										Σ
x	65.	66.	67.	67.	68.	69.	71.	73.		546.
y	67.	68.	64.	68.	72.	70.	69.	70.		548.
x·y	4355.00	4488.00	4288.00	4556.00	4896.00	4830.00	4899.00	5110.00		37422.00
x²	4225.00	4356.00	4489.00	4489.00	4624.00	4761.00	5041.00	5329.00		37314.00
Results	n	8	b	0.4242	a	39.5455	y = 39.5455 + 0.4242x			
Estimate	x	72	y at x	70.0909						

Q3. Fit y = a + bx; difference between actual y and fitted y at x = 5										Σ
x	1.	2.	3.	4.	5.	6.	7.	8.	9.	45.
y	2.	6.	7.	8.	10.	11.	11.	10.	9.	74.
x·y	2.00	12.00	21.00	32.00	50.00	66.00	77.00	80.00	81.00	421.00
x²	1.00	4.00	9.00	16.00	25.00	36.00	49.00	64.00	81.00	285.00
Results	n	9	b	0.8500	a	3.9722	y = 3.9722 + 0.8500x			
At x = 5	x	5	y fitted	8.2222	y actual	10.00	Diff.	1.7778		

Q4. Tensile test: linear relation y in terms of x (load vs elongation)										Σ
x	1.	2.	3.	4.	5.					15.
y	35.	68.	100.	138.	170.					511.
x·y	35.00	136.00	300.00	552.00	850.00					1873.00
x²	1.00	4.00	9.00	16.00	25.00					55.00
Results	n	5	b	34.0000	a	0.2000	y = 0.2000 + 34.0000x			

Q5. Fit straight line y = a + bx										Σ
x	1.	1.5	2.	2.5	3.	3.5	4.			17.5
y	1.1	1.3	1.6	2.	2.7	3.4	4.1			16.2
x·y	1.10	1.95	3.20	5.00	8.10	11.90	16.40			47.65
x²	1.00	2.25	4.00	6.25	9.00	12.25	16.00			50.75
Results	n	7	b	1.0214	a	-0.2393	y = -0.2393 + 1.0214x			

Q6. Fit y = a + bx + cx²										Σ
x	1.	2.	3.	4.	5.					15.
y	35.	68.	100.	138.	170.					511.
x²	1	4	9	16	25					55
x³	1	8	27	64	125					225
x⁴	1	16	81	256	625					979
x·y	35	136	300	552	850					1873
x²·y	35	272	900	2208	4250					7665
Results	n	5	a	2.2000	b	32.2857	c	0.2857		
Curve	y = 2.2000 + 32.2857x + 0.2857x²									

Q7. Fit second-degree parabola; estimate y at x = 6										Σ
x	1.	2.	3.	4.	5.					15.
y	5.	12.	26.	60.	97.					200.
x²	1	4	9	16	25					55
x³	1	8	27	64	125					225
x⁴	1	16	81	256	625					979
x·y	5	24	78	240	485					832
x²·y	5	48	234	960	2425					3672
Results	n	5	a	10.4000	b	-11.0857	c	5.7143		
Curve	y = 10.4000 - 11.0857x + 5.7143x²					x	6	y at x	149.6000	

Q8. Fit y = a·e^(bx) [ln y = ln a + bx]										Σ
x	0.	1.	2.	3.	4.					10.
y	1.	1.8	1.3	2.5	6.3					12.9
Y = ln y	0.0000	0.5878	0.2624	0.9163	1.8405					3.6070
x·Y	0.0000	0.5878	0.5247	2.7489	7.3622					11.2236
x²	0.0000	1.0000	4.0000	9.0000	16.0000					30.0000
Results	n	5	slope	0.4010	ln a	-0.0805	a	0.9226	b	0.4010
Curve	y = 0.9226 e^(0.4010x)									

Q9. Fit y = a·x^b [ln y = ln a + b·ln x]										Σ
x	1.	1.5	2.	2.5	3.	3.5	4.			17.5
y	1.1	1.3	1.6	2.6	2.7	3.4	4.1			16.8
X = ln x	0.0000	0.4055	0.6931	0.9163	1.0986	1.2528	1.3863			5.7526
Y = ln y	0.0953	0.2624	0.4700	0.9555	0.9933	1.2238	1.4110			5.4112
X·Y	0.0000	0.1064	0.3258	0.8755	1.0912	1.5331	1.9560			5.8880
X²	0.0000	0.1644	0.4805	0.8396	1.2069	1.5694	1.9218			6.1826
Results	n	7	slope	0.9903	ln a	-0.0408	a	0.9600	b	0.9903
Curve	y = 0.9600 x^(0.9903)									

Q10. Fit y = a·b^x [ln y = ln a + x·ln b]										Σ
x	0.	1.	2.	3.	4.					10.
y	1.	1.	1.3	2.5	6.3					12.1
Y = ln y	0.0000	0.0000	0.2624	0.9163	1.8405					3.0192
x·Y	0.0000	0.0000	0.5247	2.7489	7.3622					10.6358
x²	0.0000	1.0000	4.0000	9.0000	16.0000					30.0000
Results	n	5	slope	0.4597	ln a	-0.3156	a	0.7293	b	1.5837
Curve	y = 0.7293 (1.5837)^x									